

Fit-for-Purpose Webcam Eye Tracking for Scalable Reading Analytics

A Pilot Study with Master’s Students

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Abstract

Question. Can ordinary webcams support affordable reading analytics that explain students’ reading processes beyond quiz scores?

Study. Twenty master’s students read ten texts, answered three questions per text and provided 200 survey records. We screened eight webcam configurations.

Finding. All configurations met the hardware screen. Students reported varied strategies. Across conditions, many quizzes were perfect. Strategy recognition needs validation.

The classroom scenario

20 STUDENTS × 10 TEXTS = 200 SURVEY RECORDS

Master’s students at **Gisma University of Applied Sciences, Potsdam campus** read about multi-agent systems on their own laptops after a lecture.

- 1

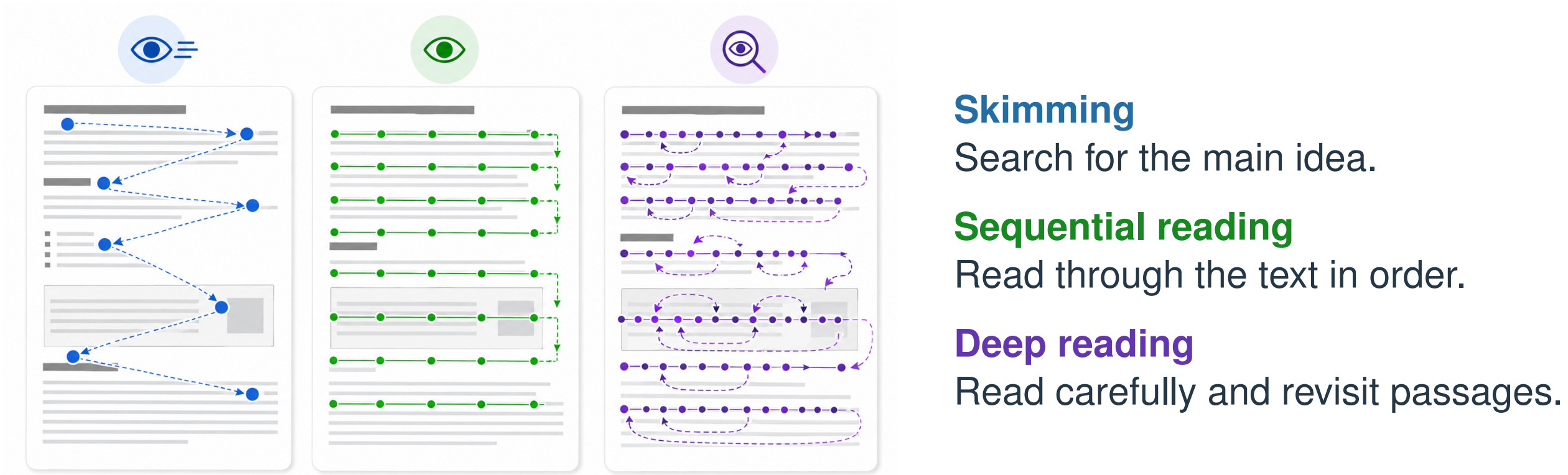
Read at your own pace
Example topic: how several AI agents work together.
- 2

Answer three questions
One easy, one intermediate and one hard question per text.
- 3

Report the experience
Select a strategy; rate difficulty, effort, understanding, familiarity and confidence.

The webcam application logged events during the task. Survey selections supply the strategy labels re-reported here.

Conceptual illustration: reading patterns

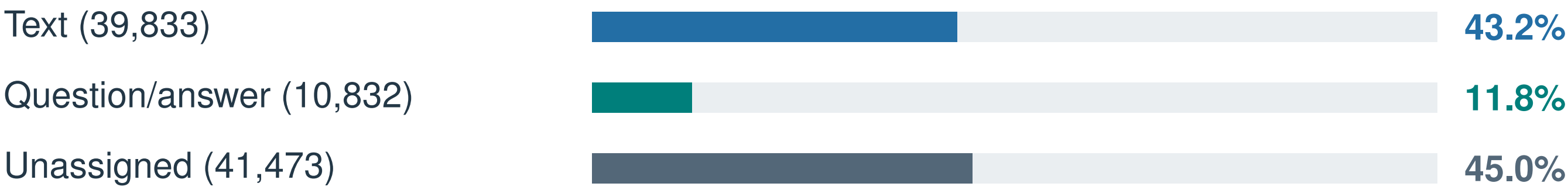


Examples from the slides; no participant’s measured gaze path is shown.

What the webcam logs establish

APPLICATION LOGS: ALL READING CONDITIONS

92,138 logged events across 60 app sessions. Each event has a time interval and have a rectangle-assigned area ID.



55.0% received an area label: 50,665 labelled events divided by 92,138 events. This measures assignment coverage, not location accuracy or reading time. For unassigned events, the export cannot distinguish gaze outside labelled areas from tracking or mapping problems.

What an area label means

An **area of interest (AOI)** is part of the displayed material. Here, word labels are grouped into source text or questions/answers.

What remains unvalidated

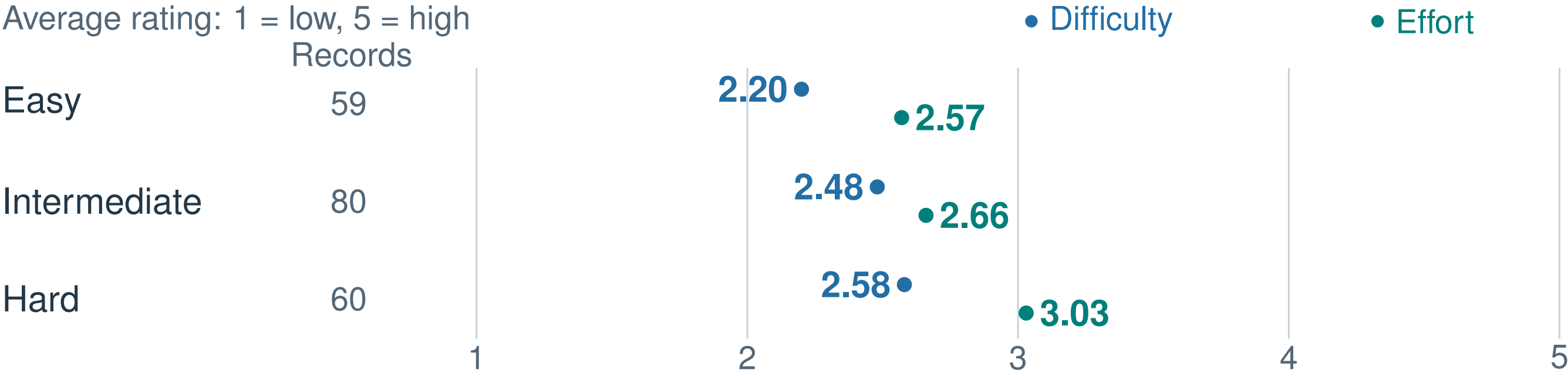
No camera calibration was performed.

The exports lack raw gaze coordinates, validation errors and clear initial-reading boundaries. They cannot establish exact skipped words, rereading movements or accurate strategy recognition.

What students reported

NORMAL READING: 20 STUDENTS, 200 RECORDS

Harder texts received slightly higher effort ratings.



Effort: 2.57 (easy) to 3.03 (hard), a +0.46-point difference. Each average is the sum of ratings divided by available responses. Groups use 199 records; one lacked a text-difficulty label. Descriptive averages, not causal effects.

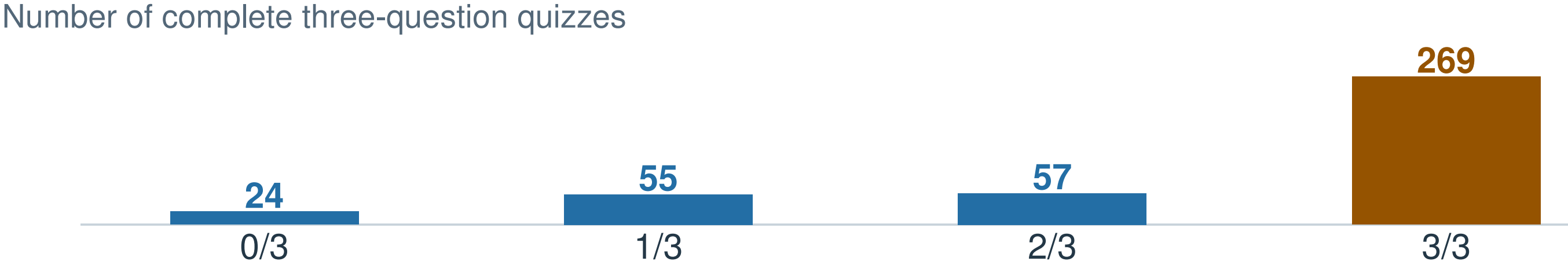
Reported strategies varied across texts



Many quizzes reached the maximum score

APPLICATION QUIZZES: ALL READING CONDITIONS

269 of 405 complete quizzes scored 3/3: 66.4%.



Many quizzes reached the maximum score, the **ceiling**, limiting comparisons between learners. Each quiz comprises three scored questions for one text within an app session. This sample is not the 200 survey records.

Eight webcam configurations

HARDWARE SCREEN: 8 DISTINCT CONFIGURATIONS

20–30 FPS; 1280×720 to 2560×1440 pixels. All configurations met the study’s minimum hardware criteria.

Camera	Device / OS	Pixels	FPS
HP HD Camera	HP / Win 11	1280×720	25
Integrated Camera A	Acer/Asus / Win 11	1280×720	30
Integrated Camera B	Acer/Asus / Win 11	1280×720	29
Integrated Webcam	Dell / Win 11	2560×1440	30
Samsung Camera Front	Motorola / Win 11	1920×1080	29
MacBook Air Camera	MacBook / macOS	1920×1080	28
USB2.0 HD UVC Webcam	Legion / Win 11	1280×720	20
ASUS HD webcam	Acer/Asus / Win 11	1280×720	30

Criteria used in this study

Minimum: 1280×720 pixels and 20 FPS.
Preferred: 1920×1080 and 25 FPS; met by the Dell, Samsung and MacBook configurations.

FPS means frames per second. The Legion USB webcam was slowest: 20 FPS, one frame every 50 ms. Hardware screening does not validate gaze accuracy. Names follow the source report; A/B distinguish repeated labels. Only two camera records were fully linked to gaze logs. Webcam-test FPS is not the app’s effective sampling rate.

Next question: does validated gaze add useful information for teachers?

1. Validate and link.

Check gaze against known targets. Use one participant ID and explicit reading/visibility events.

2. Measure learning better.

Use questions that separate levels of understanding, including de-layed and transfer tasks.

3. Test strategy agreement.

Compare independent gaze-based labels with matched surveys. Test future models on new students.

Sources: GitHub repos <https://github.com/duongtrung/DELFI2026>

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